Lab 4: Functional Dependencies

Reda Lakhlouf

October 2026

Part 1 gives the FDs from the keys. Part 2 gives the FDs from the foreign keys. The code in brackets is the requirement from Lab 1. "Key only" means the relation has no other FD. The right sides are grouped with the union rule: if $X \to Y$ and $X \to Z$, then $X \to YZ$.

1 Part 1: FDs from the keys

1.1 People

- Person.pid → firstName, lastName, email, affiliation, startDate, endDate (PE1)
- Student.pid → program (PE3)
- Employee.pid → phone, office, supervisorPid (PE4, PE5: the dean reports to nobody, so supervisorPid can be NULL)
- Faculty.pid, Administrative.pid, Technical.pid → position (PE4)
- Academic(pid), NonAcademic(pid), Advises(academicPid, studentPid): key only

Remarks:

- supervisorPid does not give pid, because a supervisor has many subordinates.
- In Advises, one attribute does not give the other (PE6).
- "No self-report" and "report to the dean" (PE5) are constraints, not FDs.

1.2 Laboratories, Projects, and Budgets

- Laboratory.labId → name, building, roomNumber, discipline, supervisorPid (LA1, LA4)
- ResearchProject.code → title, startDate, endDate, status (PR1)
- Participates.(pid, projectCode) → role (PR3, PR4)
- Budget.budgetLine → amountGranted, amountDisbursed, startDate, endDate, managerPid (BD1, BD2)
- Attached, FundsPrj, FundsLab: key only (LA3, BD3)

Remarks:

- In Participates, pid alone gives nothing (PR2) and projectCode alone gives nothing (PR3). Both are in the key.
- "One PI for each project" (PR3) is (projectCode, role = PI) → pid. It is true for one value of role only, so it is a constraint, not an FD.
- supervisorPid does not give labId (LA4), and managerPid does not give budgetLine (BD2).

1.3 Equipment and Certifications

- EquipmentModel.modelId → commercialName, manufacturer, category, requiredEnvironment, trainingMandatory (EQ2)
- EquipmentUnit.serialNo → acquisitionDate, purchaseCost, status, portable, modelId, labId (EQ3–EQ5)
- Certification.code → title, issuingAuthority, validityPeriod, safetyLevel (CE2)
- Holds.(pid, certCode) → issueDate, expirationDate, grade (CE4, CE5)
- Requires(modelId, certCode): key only

Remarks:

- modelId and labId do not give serialNo, because many units have the same model or lab.
- If the model attributes were in EquipmentUnit, modelId would give them but it is not a key. This is the same problem as $R \to W$ in Hourly_Emps. EQ1 (model and unit separate) avoids it.
- In Holds, (pid, certCode) → pid is true by reflexivity, but trivial, so I do not list it.

1.4 Reservations

- Reservation.resId → submissionTS, plannedStart, plannedEnd, purpose, status, madeByPid, projectCode, approverPid (RE1–RE3; approverPid can be NULL)
- Reserves(resId, serialNo): key only

Remarks:

- madeByPid does not give projectCode, and approverPid does not give resId (RE3).
- The rules RE1, CE7, EQ7 and RE4 depend on other tables or on time, so they are constraints.

1.5 Maintenance and Calibration

- Maintenance.(serialNo, startTS) → endTS, type, description, cost, outcome, technicalPid (MA2, MA3)
- CalibrationRecord.(serialNo, calibDate) → calibrationType, result, nextDueDate, remarks (CA2–CA4)

Remark: A unit has many maintenances and calibrations (MA1), so serialNo alone is not enough, and startTS or calibDate alone is not enough.

1.6 Consumables, Suppliers, and Stock

- Consumable.consId → name, unitOfMeasure, hazardLevel, reorderThreshold (CO2)
- Supplier.suppId → name, contactEmail, phone (CO3)
- Supplies.(suppId, consId, labId) → unitPrice (CO4)
- Stocks.(labId, consId) → quantityOnHand, lastRestockDate, storageCondition, technicalPid, monitoringSince (CO6–CO8)
- Consumes.(resId, labId, consId) → quantityUsed (CO9)

Remarks:

- In Supplies, (suppId, consId) does not give unitPrice (CO4), so the three attributes are the key.
- In Stocks, labId alone and consId alone give nothing (CO6).
- Supplies and Stocks are separate because of CO5.

In every relation the left side of each FD is a key, so there is no FD like $R \to W$.

2 Part 2: FDs from the foreign keys

When a foreign key points to the key of another relation, it gives all the attributes of that relation. The key that contains the foreign key gives them too, by transitivity. Each attribute has the name of its relation in front, because some names are in several relations.

2.1 People

- Student.pid → Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Employee.pid → Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Academic.pid → Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- NonAcademic.pid → Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Faculty.pid → Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Administrative.pid → Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Technical.pid → Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Employee.supervisorPid → Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (all of the supervisor)

- Advises.academicPid → Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Advises.studentPid → Student.program, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate

2.2 Laboratories, Projects, and Budgets

- Laboratory.supervisorPid → Faculty.position, Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Attached.pid → Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Attached.labId → Laboratory.name, Laboratory.building, Laboratory.roomNumber, Laboratory.discipline, Laboratory.supervisorPid
- Attached.labId → Faculty.position, Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (of the supervisor of the laboratory)
- Participates.pid → Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Participates.projectCode → ResearchProject.title, ResearchProject.startDate, ResearchProject.endDate, ResearchProject.status
- Budget.managerPid → Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- FundsPrj.budgetLine → Budget.amountGranted, Budget.amountDisbursed, Budget.startDate, Budget.endDate, Budget.managerPid
- FundsPrj.budgetLine → Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (of the manager)
- FundsPrj.projectCode → ResearchProject.title, ResearchProject.startDate, ResearchProject.endDate, ResearchProject.status
- FundsLab.budgetLine → Budget.amountGranted, Budget.amountDisbursed, Budget.startDate, Budget.endDate, Budget.managerPid
- FundsLab.budgetLine → Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (of the manager)
- FundsLab.labId → Laboratory.name, Laboratory.building, Laboratory.roomNumber, Laboratory.discipline, Laboratory.supervisorPid
- FundsLab.labId → Faculty.position, Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (of the supervisor of the laboratory)

2.3 Equipment and Certifications

- EquipmentUnit.modelId → EquipmentModel.commercialName, EquipmentModel.manufacturer, EquipmentModel.category, EquipmentModel.requiredEnvironment, EquipmentModel.trainingMandatory
- EquipmentUnit.labId → Laboratory.name, Laboratory.building, Laboratory.roomNumber, Laboratory.discipline, Laboratory.supervisorPid
- EquipmentUnit.labId → Faculty.position, Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (of the supervisor of the laboratory)
- Requires.modelId → EquipmentModel.commercialName, EquipmentModel.manufacturer, EquipmentModel.category, EquipmentModel.requiredEnvironment, EquipmentModel.trainingMandatory
- Requires.certCode → Certification.title, Certification.issuingAuthority, Certification.validityPeriod, Certification.safetyLevel
- Holds.pid → Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Holds.certCode → Certification.title, Certification.issuingAuthority, Certification.validityPeriod, Certification.safetyLevel

2.4 Reservations

- Reservation.madeByPid → Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Reservation.approverPid → Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (only when approverPid is not NULL)
- Reservation.projectCode → ResearchProject.title, ResearchProject.startDate, ResearchProject.endDate, ResearchProject.status
- Reserves.resId → Reservation.submissionTS, Reservation.plannedStart, Reservation.plannedEnd, Reservation.purpose, Reservation.status, Reservation.madeByPid, Reservation.projectCode, Reservation.approverPid
- Reserves.resId → Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (of the person who made it, and of the approver), and → ResearchProject.title, ResearchProject.startDate, ResearchProject.endDate, ResearchProject.status
- Reserves.serialNo → EquipmentUnit.acquisitionDate, EquipmentUnit.purchaseCost, EquipmentUnit.status, EquipmentUnit.portable, EquipmentUnit.modelId, EquipmentUnit.labId
- Reserves.serialNo → EquipmentModel.commercialName, EquipmentModel.manufacturer, EquipmentModel.category, EquipmentModel.requiredEnvironment, EquipmentModel.trainingMandatory, and → Laboratory.name, Laboratory.building, Laboratory.roomNumber, Laboratory.discipline, Laboratory.supervisorPid

2.5 Maintenance and Calibration

- Maintenance.serialNo → EquipmentUnit.acquisitionDate, EquipmentUnit.purchaseCost, EquipmentUnit.status, EquipmentUnit.portable, EquipmentUnit.modelId, EquipmentUnit.labId

- Maintenance.serialNo → EquipmentModel.commercialName, EquipmentModel.manufacturer, EquipmentModel.category, EquipmentModel.requiredEnvironment, EquipmentModel.trainingMandatory, and → Laboratory.name, Laboratory.building, Laboratory.roomNumber, Laboratory.discipline, Laboratory.supervisorPid
- Maintenance.technicalPid → Technical.position, Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- CalibrationRecord.serialNo → EquipmentUnit.acquisitionDate, EquipmentUnit.purchaseCost, EquipmentUnit.status, EquipmentUnit.portable, EquipmentUnit.modelId, EquipmentUnit.labId
- CalibrationRecord.serialNo → EquipmentModel.commercialName, EquipmentModel.manufacturer, EquipmentModel.category, EquipmentModel.requiredEnvironment, EquipmentModel.trainingMandatory, and → Laboratory.name, Laboratory.building, Laboratory.roomNumber, Laboratory.discipline, Laboratory.supervisorPid

2.6 Consumables, Suppliers, and Stock

- Supplies.suppId → Supplier.name, Supplier.contactEmail, Supplier.phone
- Supplies.consId → Consumable.name, Consumable.unitOfMeasure, Consumable.hazardLevel, Consumable.reorderThreshold
- Supplies.labId → Laboratory.name, Laboratory.building, Laboratory.roomNumber, Laboratory.discipline, Laboratory.supervisorPid
- Supplies.labId → Faculty.position, Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (of the supervisor of the laboratory)
- Stocks.labId → Laboratory.name, Laboratory.building, Laboratory.roomNumber, Laboratory.discipline, Laboratory.supervisorPid
- Stocks.labId → Faculty.position, Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (of the supervisor of the laboratory)
- Stocks.consId → Consumable.name, Consumable.unitOfMeasure, Consumable.hazardLevel, Consumable.reorderThreshold
- Stocks.technicalPid → Technical.position, Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate
- Consumes.resId → Reservation.submissionTS, Reservation.plannedStart, Reservation.plannedEnd, Reservation.purpose, Reservation.status, Reservation.madeByPid, Reservation.projectCode, Reservation.approverPid
- Consumes.resId → Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (of the person who made it, and of the approver), and → ResearchProject.title, ResearchProject.startDate, ResearchProject.endDate, ResearchProject.status
- Consumes.(labId, consId) → Stocks.quantityOnHand, Stocks.lastRestockDate, Stocks.storageCondition, Stocks.technicalPid, Stocks.monitoringSince
- Consumes.(labId, consId) → Consumable.name, Consumable.unitOfMeasure, Consumable.hazardLevel, Consumable.reorderThreshold, and → Laboratory.name, Laboratory.building, Laboratory.roomNumber, Laboratory.discipline, Laboratory.supervisorPid

- Consumes.(labId, consId) → Technical.position, Employee.phone, Employee.office, Employee.supervisorPid, Person.firstName, Person.lastName, Person.email, Person.affiliation, Person.startDate, Person.endDate (of the technician who monitors the stock)

3 Hypotheses

The requirements do not give these FDs. They are true only if the assumption is true.

- If an email belongs to one person only, then Person.email → pid. Email is then also a candidate key.
- If a supplier's contact email is unique, then Supplier.contactEmail → suppId.
- If a phone number or an office is never shared, then Employee.phone → pid and Employee.office → pid.
- If a room has at most one laboratory, then Laboratory.(building, roomNumber) → labId.
- If a project title is unique, then ResearchProject.title → code.
- If the expiration date is always the issue date plus the validity period, then Holds.(certCode, issueDate) → expirationDate. Here (certCode, issueDate) is not a key of Holds, so the expiration date is repeated.

The first five give a new candidate key. Only the last one has a left side that is not a key.

4 Report

4.1 Role of each member

I did all the tasks myself.

1. First, I reviewed Lecture 4: what an FD is, keys, Armstrong's axioms and attribute closure.
2. Second, I went back to Lab 1 and Lab 2 to see how the relations are linked with primary keys and foreign keys.
3. Third, I wrote the FDs from the keys (Part 1), then the FDs from the foreign keys (Part 2).

4.2 Key challenges

- The schema has thirty relations, so there are many dependencies. I made two parts, keys and foreign keys. In each part I followed the sections of Lab 2 and checked Lab 1 for the cardinalities.
- My first draft put attributes of several relations in one FD, for example Person attributes in an FD of Student. I fixed it: Part 1 stays in one relation, and Part 2 has the FDs between relations.
- Some rules look like FDs but are not, for example "exactly one PI" or "at least one advisor". I wrote them in the remarks.

4.3 Justification of the choices

- For each relation I started with the key, because a key gives all the other attributes.
- In Lab 2, a relationship with "at most one" or "exactly one" on one side became a foreign key in a table (Supervises, Manages, InstanceOf, LocatedIn, MadeBy, For, Approves, DoneBy, Monitors, ReportsTo). So the key gives this foreign key, for example labId $\rightarrow$ supervisorPid. For "at most one", the foreign key can be NULL.
- M:N relationships without attributes have only the key, so they are "key only".
- For the composite keys (Supplies, Stocks, Maintenance, CalibrationRecord, Participates), I computed the closure of each part of the key. None gives the other attributes, so there is no partial dependency.
- The FDs from the foreign keys come from transitivity. If the referenced relation has foreign keys too, the chain continues, so I wrote the second level.
- The model and unit split in Lab 1 is a decomposition. Without it, modelId would give the model attributes and it is not a key.
- Every FD comes from a requirement, not from sample data. An instance can show that an FD is false, but it cannot show that it is true.

4.4 Assumptions

The FDs that need an assumption are in the Hypotheses section, each written as "if this is assumed, this FD holds". Without them, only the FDs of Part 1 and Part 2 are valid.